

JOANNE HOAR

Senior Software Developer

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Hanna, Alberta, Canada • Remote

Summary

Canadian Software Developer with 20 years of experience in video game programming, scientific modelling, and cross-platform development. Expertise in multiple programming languages, game engines, and SDKs. Proven track record in leading teams, mentoring developers, and building corporate culture. Strong background in C++, Unreal Engine, iOS, and Android using agile methodologies and CI/CD. Passion for algorithmic correctness, mathematics, artificial intelligence, and nature.

Core Competencies

- C++, C#, Objective-C, Swift, Kotlin
- Unreal Engine & Game SDK Integration
- iOS & Android Native Development
- Augmented Reality (ARKit) & OpenGL Graphics
- CI/CD, TDD & Agile Methodologies
- ASP .Net Core, React, Node.js, JavaScript
- Team Leadership & Developer Mentorship
- Technical Curriculum Design

Professional Experience

Technology Instructor

Jan 2025 – Present

Robertson College (contract) — Online Instructor-Guided

- Instructed 20 course offerings across 12 subjects for **Full Stack Development** and **Data Analytics** diploma programs.
- Courses: C# Programming, .NET, ASP .NET/Core, Object-oriented Programming (Java), Front End Frameworks, Intro to JavaScript, Web Development Basics, Data Collection, Testing & QA, Version Control with Git & GitHub, Introduction to IT.
- Deliver 3+ synchronous live sessions per week per course, including live coding and real-time student debugging.
- Maintain a GitHub repository of open-source course materials; archive session recordings for asynchronous access.
- Maintained average student satisfaction rating of **4.5/5.0** across evaluated courses; students cited approachability, real-world teaching style, and responsiveness.

C#, .NET, ASP .NET Core, Java, JavaScript, React, Git, GitHub, Azure, SQL, Testing & QA

Senior Software Developer

2022 – 2025

bitHeads Inc. — brainCloud BaaS

- Developed and maintained cross-platform client SDKs for **brainCloud BaaS**, enabling gamification, leaderboards, matchmaking, lobbies, real-time multiplayer, and cloud code for **50+ published games** across **12 platforms** (PlayStation, Xbox, Nintendo Switch, Steam, iOS, Android, Oculus, and more).
- Primary focus: C++ client SDK, Unreal Engine plugin, iOS (Objective-C/Swift), Android Native (Java/Kotlin), and PlayStation; reviewed and debugged across all supported platforms and languages.
- Designed and built demonstration applications and widget libraries for every supported client platform — UI frameworks spanned **Dear ImGui** (C++), Unreal Engine UMG, SwiftUI/UIKit (iOS), Android Views (Java/Kotlin), Unity (C#), and PlayStation SDK.
- Built and maintained Unreal Engine sample projects (Space Shooter, Target Practice, Tappy Chicken, Relay Blast, dedicated server) to onboard client studios.
- Supported integration for notable titles including *SUPERHOT VR*, *Tony Hawk's Skate Jam*, *Crypt of the NecroDancer*, and *Invizimals* (PlayStation Mobile).
- Hired in part for educational background — role required teaching clients one-on-one, strong documentation skills, and developing academic examples and demonstrations.
- Engineered and maintained CI/CD and TDD pipelines; responded directly to customer issues and feature requests from AAA and Indie studio clients.

Unreal Engine, C++, Objective-C, Swift, iOS, Android Native, Java, Kotlin, PlayStation, Nintendo Switch, Xbox, Node.js, Groovy, C# (Unity, Godot), Shell Script, macOS, Windows, Linux

Co-founder / Lead Developer / Instructor

Silicon Hanna Inc. (self-employed; founded as SuRJE Software Solutions Inc.; concurrent with BJ Pipeline 2005–08 and Fekete Associates Inc. 2008–2011)

- Designed, coded, and published eleven games to commercial app stores for mobile devices across iOS and Android.
- Developed curriculum and instructed intermediate C++ programming for the **University of Calgary Department of Continuing Education**; received positive student feedback.
- Explored cutting-edge device technologies and SDKs including ARKit augmented reality, Chipmunk physics, and Unreal Engine.
- Technical writing, web development, Google Maps and Firebase integration, and server administration.

Unreal Engine, Objective-C / Swift, iOS, Android Native, Kotlin, ARKit, C++, OpenGL, PHP, SQL, macOS, Windows, Linux

Tools Programmer

2008 – 2011

Fekete Associates Inc. (S&P Global)

- Developed tools using mathematical models to maximize production and improve efficiency for major oil & gas corporations.
- Contributed to the shared MFC-based UI tools library used as the consistent component foundation across all internal development projects.
- Led research on a new build system that reduced build times by 40%.

- Designed and delivered a voluntary OpenGL elective for colleagues — presentations, sample code, and weekly assignments for both beginner and senior developers; received positive response from students and management.

C++, MFC, Windows

Intermediate Software Engineer

2005 – 2008

BJ Pipeline Inspection Services Inc. (Baker Hughes)

- Visualized in-line inspection device data using OpenGL graphics for **Vectra** and **Gemini** 3D analysis products used by pipeline analysts and engineers.
- Implemented neural network algorithms for automated anomaly detection.
- Ported an Excel-based inspection reporting workflow to a full LAMP web application with interactive JavaScript forms, dynamic charts, and interactive geographic maps — replacing a static spreadsheet process with a live, accessible tool for non-technical stakeholders.
- Improved low-level C++ algorithms for high-performance data preprocessing.

C++, C, OpenGL, PHP, JavaScript, SQL, Neural Networks, LAMP, Borland, Windows, Linux

Earlier Career Highlights

Software Researcher / Lead Developer

2005 – 2006

PlayStarMusic Corp. (concurrent with BJ Pipeline)

- Designed and built the end-user interface for a media streaming storefront — browsing, playback controls, purchase flow, and seamless DRM integration that kept the listening experience uninterrupted.
- Lead developer and system administrator for the full platform stack; developed a prototype for a competitive search engine.

C++, PHP, JavaScript, .NET, Windows, Linux

Junior Software Engineer

2004 – 2005

Codefast.com

- Primary mandate: building graphical user interfaces for existing CLI tools — translating command-line workflows into accessible desktop applications.
- Ported applications across GUI libraries (Qt, Wx); developed user interfaces for software lifecycle automation tooling.

C++, Qt, Wx, Shell Script, Windows, Unix

Research Assistant

2001 – 2004

VLab Computer Graphics Group / Artificial Intelligence Research Hub

- Developed graphics software tools using fractal mathematics and biological modelling.
- Managed network setup and code base for distributed computing infrastructure.
- Participated in interdepartmental projects and published five research papers.
- Served as teaching assistant for first-year and advanced third-year courses — supplemented instructor lesson plans with independent presentations and code/math walk-throughs.

C++, OpenGL, Solaris, macOS, Unix, Linux, PHP, MySQL, Qt

Education

- **M.Sc. Computer Science** – University of Calgary 2004
Thesis: Agent-based Development of Natural Transportation Networks
- **B.Sc. Computer Science** – University of Calgary 2001

Selected Publications

- Hoar, J., et al. "Bacterial Chemotaxis in Silico," *Australian Conference on Artificial Life (ACAL)*, 2003 *Best Student Paper Award*
- Hoar, R., Jacob, C. "The application of artificial intelligence to transportation system design," *XRDS: Crossroads, ACM Magazine for Students*, 2003

Honors & Awards

- Departmental Nomination: Governor General's Gold Medal, Faculty of Graduate Studies 2005
- Postgraduate Scholarship NSERC D3, Natural Sciences and Engineering Research Council of Canada 2005
- Graduate Student Scholarship, Alberta Learning 2004
- Best Student Paper Award (Co-author), ACAL 2003
- CPSC Departmental Research Award, University of Calgary 2003

Published Games

Silicon Hanna Inc. (founded as SuRJE Software Solutions Inc.) — 11 titles, 2008–2022:

- Ragin' Bull – iOS, Android (2020)
- AR Picture Portal – iOS (2018)
- Fragment Reality – iOS (2018)
- IceCream AR – iOS (2018)
- Jewel Slide – iOS, Android (2017, 2019)
- Word Slide & Find – iOS (2017)
- A Word Theme – iOS, Android (2017, 2019)
- CryptoQuip – iOS, Android (2011, 2019)
- Rafter's Loot – iOS (2011)
- Star Chase – iOS (2008)
- Bitsy – iOS (2008)

Community Outreach & Leadership

- Treasurer, Hanna & District Minor Soccer Association 2025–Present
- Vice President, Hanna Figure Skating Club 2024–Present
- Costume Coordinator, Stage Hanna Junior 2024
- Women of bitHeads and Cookbook Committee, bitHeads 2024
- Team Manager / Assistant Coach, U11 Soccer 2024
- Dryland Training Coordinator, Hanna Figure Skating Club 2022–2024
- 4-H Public Speaking Judge, District Level Competition 2022
- Computer Programming for Kids, J.C. Charyk School 2019–2021
- Reading Program, Our Lady of Fatima School 2016–2018
- Preschool Gym Leader, YMCA 2013–2014
- OpenGL Elective Instructor, Fekete Associates 2010
- Calgary Corporate Challenge 2007–2008
- Ladies Luncheon Club and Golf Tournament Coordinator, BJ Pipeline Inspection 2006–2008
- SCIBerMENTOR Correspondent / Mentor 2002–2005
- President, Computer Science Graduate Society 2002–2003
- VP Academic, Computer Science Undergraduate Society 2001

Technical Skills

Languages: C++, C#, JavaScript, Java, Objective-C, Swift, Kotlin, PHP, SQL

Game & Graphics: Unreal Engine, OpenGL, ARKit, Augmented Reality, Real-time Multiplayer, Game SDK Integration

Platforms & Frameworks: iOS, Android Native, ASP .Net Core, React, Node.js, Qt, MFC

Tools & Practices: GitHub, Azure, CI/CD, TDD, Shell Script, Agile Methodologies

Languages (Human): English (native), French (intermediate), Spanish (rudimentary)

Work Samples

 github.com/xcodegirl/career – xcodegirl-work-samples.md